



CS231. Nhập môn Thị giác máy tính

Bài 00. Giới thiệu môn học



Thông tin chung

Tên môn học (tiếng Việt):	Nhập môn Thị giác Máy tính
Tên môn học (tiếng Anh):	Introduction to Computer Vision
Mã môn học:	CS231
Thuộc khối kiến thức:	Đại cương ☐ ; Cơ sở nhóm ngành ☐ ; Cơ sở ngành ☐ ; Chuyên ngành ☒ ; Tốt nghiệp ☐



Mục tiêu môn học

1. Làm việc ở mức độ cá nhân và cộng tác nhóm để trình bày và giải quyết một số bài toán liên quan tới Thị giác máy tính.
2. Hiểu và giải thích được các khái niệm cơ bản, thuật ngữ liên quan tới Thị giác Máy tính



Mục tiêu môn học

3. Ứng dụng các kiến thức căn bản về Thị giác máy tính để ứng dụng vào các vấn đề đơn giản trong thực tế.
4. Ứng dụng các công cụ hỗ trợ, thư viện mở về Thị giác Máy tính.



Đánh giá

A4. Cuối kỳ - LT cuối kỳ - Đồ án cuối kỳ		70 % (30% + 40%)
A3. Thực hành		30 %



Bài tập

- Mỗi buổi sẽ có bài tập.
- Tên file = Mã số học viên.
- Deadline cho bài tập là 1 tuần
- Sinh viên xong và đúng yêu cầu trong buổi học: 10 điểm
- Sinh viên có đi học nhưng chưa xong, cập nhật sau theo đúng yêu cầu trong buổi học: 9 điểm
- Sinh viên viên vắng có phép, bổ sung bài tập đúng: 9 điểm
- Sinh viên vắng không phép, bổ sung bài tập đúng: 8 điểm
- Nộp bài của bạn: 0 điểm

- Có 3 nội dung phải nộp:
- 1. Báo cáo tổng kết: Nội dung báo cáo tổng kết được trình bày trong file pdf hoặc docx, có ít nhất các nội dung:
 - Thông tin các thành viên và đánh giá mức độ hoàn thành
 - Những lý do hoặc tầm quan trọng của bài toán
 - Phát biểu bài toán (Input, Output)
 - Phương pháp giải quyết
 - Kết quả thực nghiệm (dataset, độ đo, lập trình, kết quả,...)
 - Đánh giá, nhận xét những ưu điểm và hạn chế
 - Tài liệu tham khảo

- 2. Source code:
- Các em có thể nộp link github, hoặc gdrive (lưu ý cấp quyền viewer, không cấp mà yêu cầu GV phải request thì xem như không có nộp) hoặc chép tất cả vào thư mục nộp đồ án.

3. Side trình bày: slide trình bày trên lớp

- Thông tin các thành viên
- Những lý do hoặc tầm quan trọng của bài toán
- Phát biểu bài toán (Input, Output)
- Phương pháp giải quyết
- Kết quả thực nghiệm (dataset, độ đo, lập trình, kết quả,...)
- Đánh giá, nhận xét những ưu điểm và hạn chế
- Tài liệu tham khảo

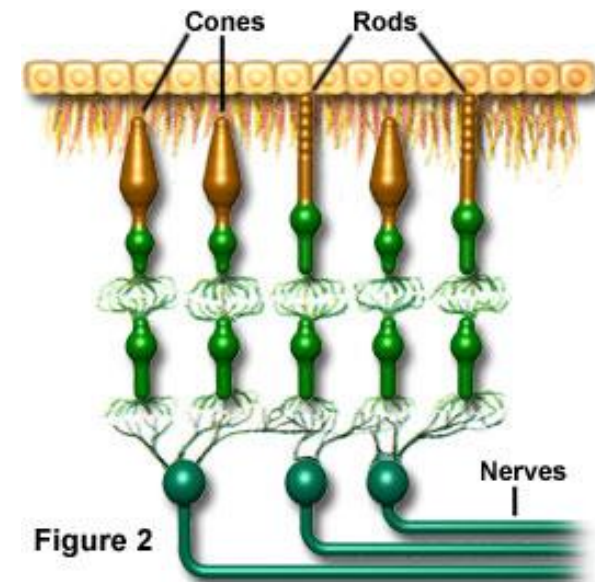
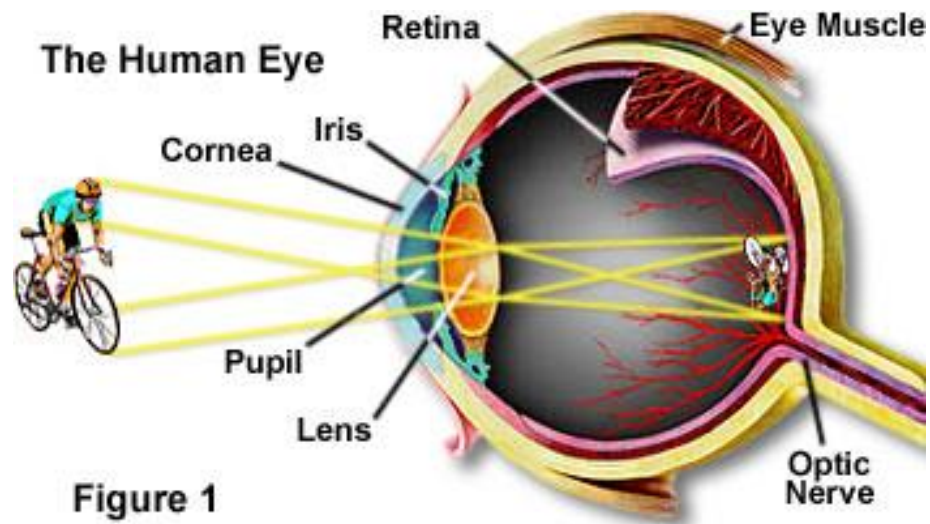


What is computer vision ?



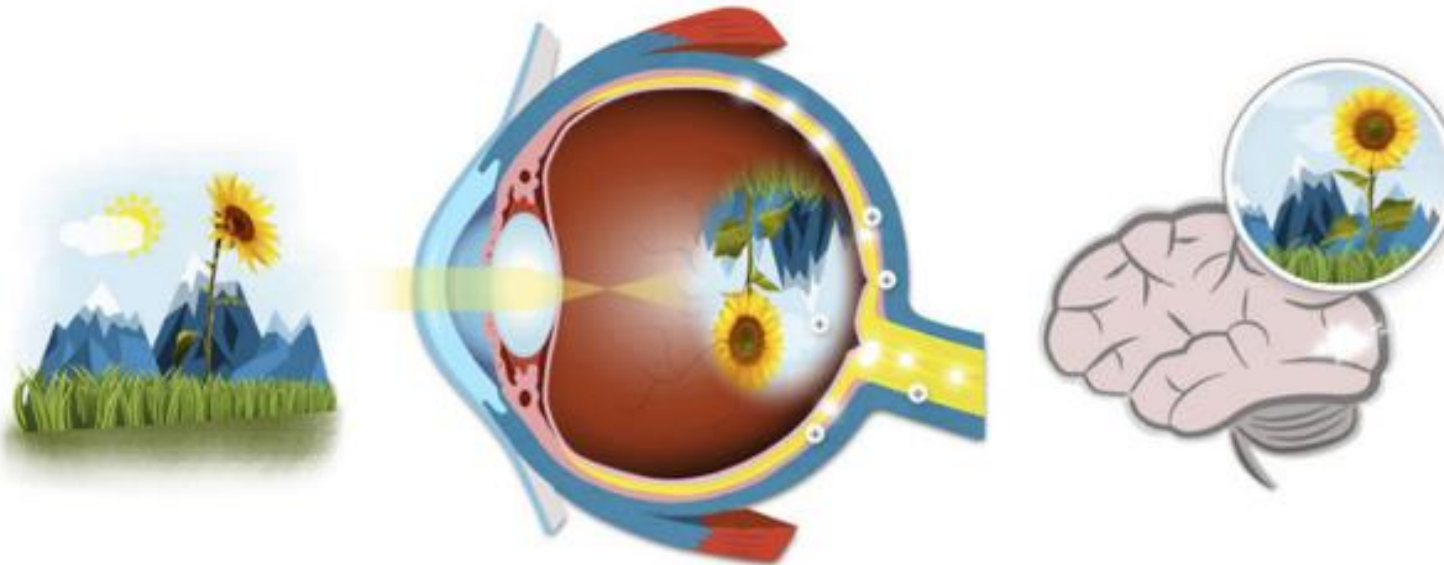
What is vision ?

- Human Vision



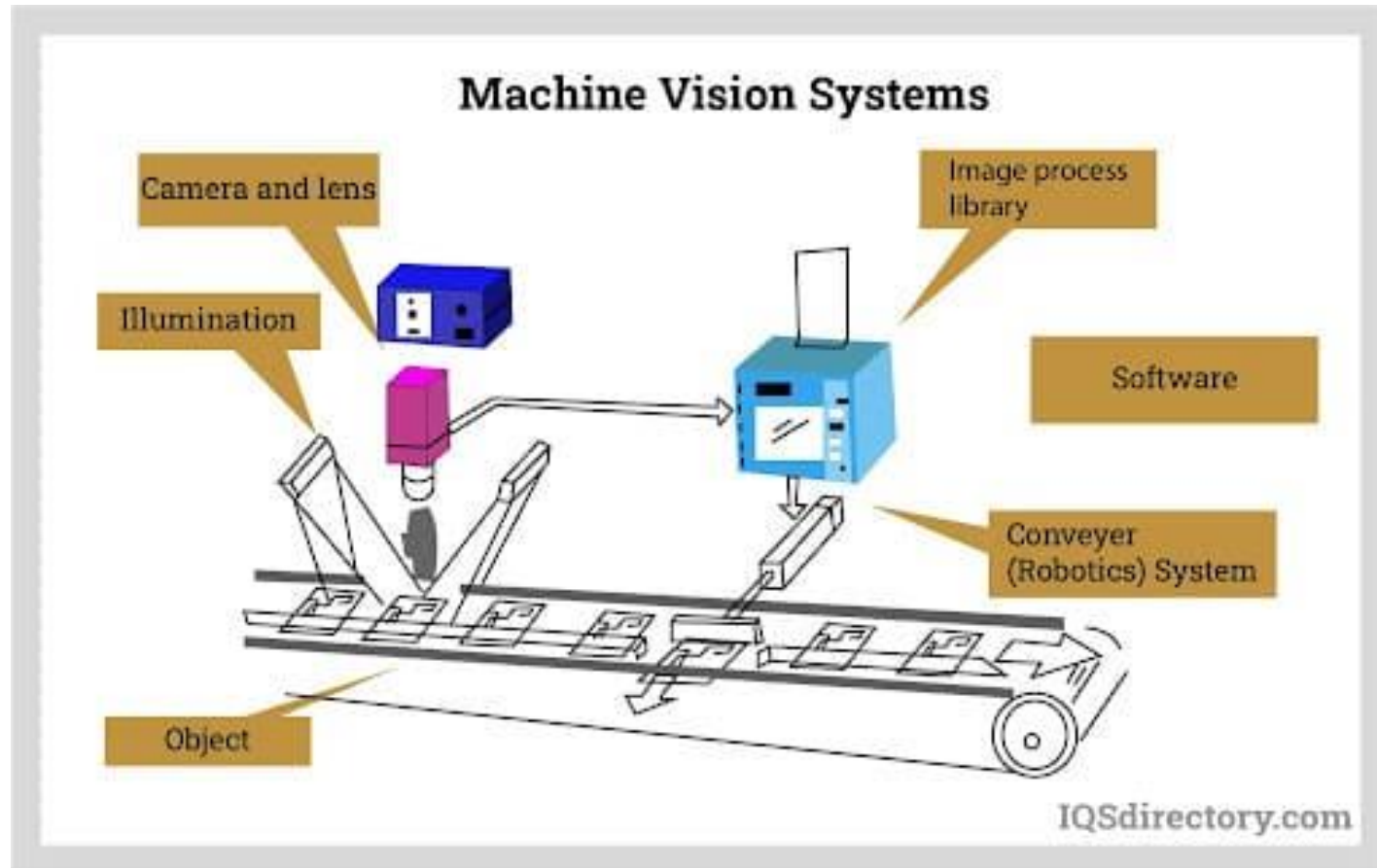
What is **vision** ?

- Human Vision



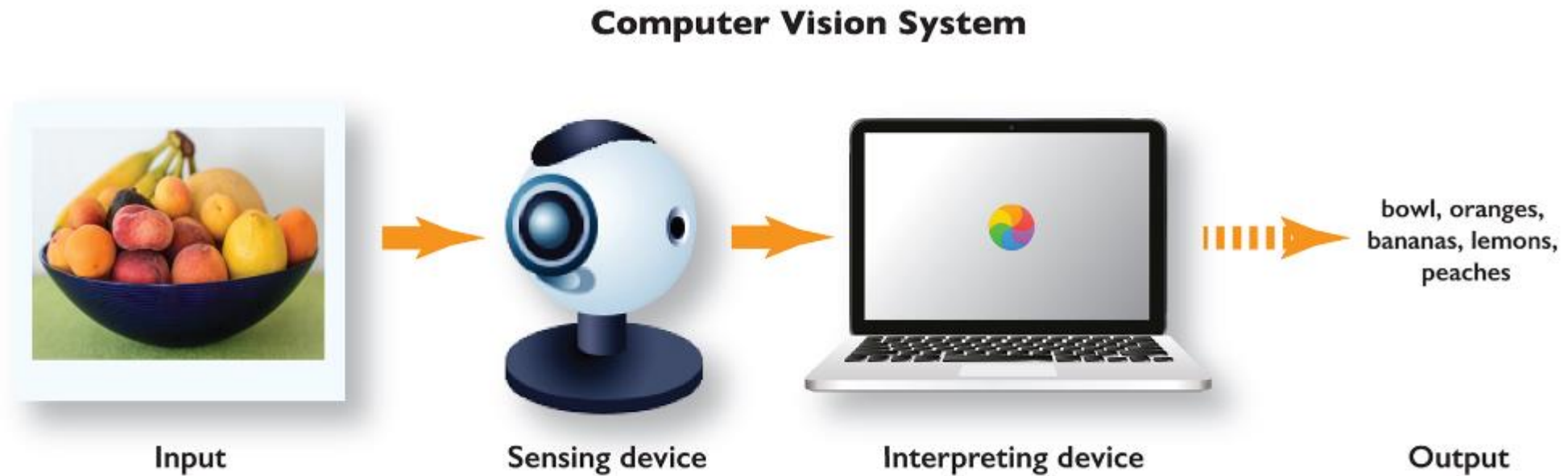
What is **vision** ?

- Machine Vision



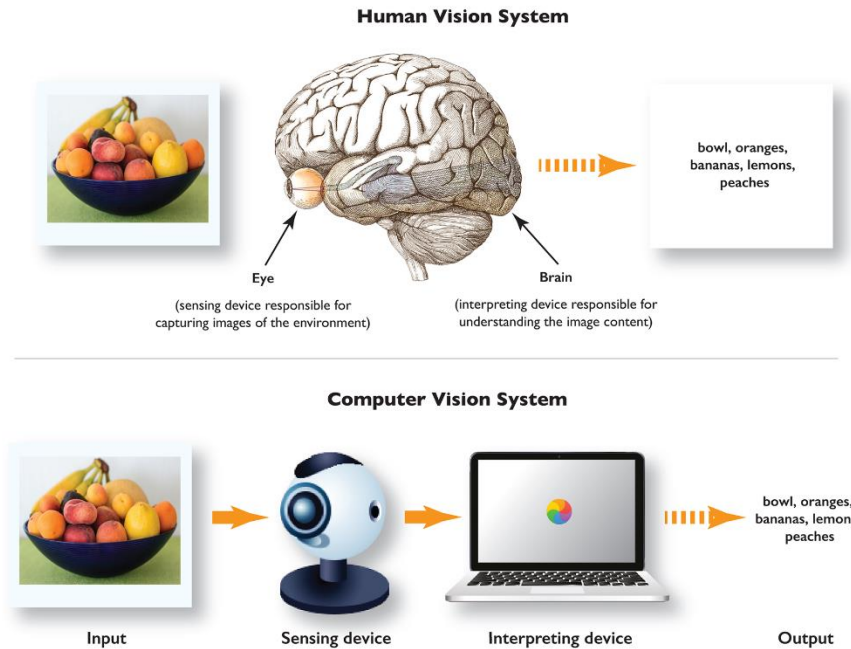
What is **vision** ?

- Computer vision



What is **vision** ?

- Human vs Computer vision





What is computer vision ?

- <https://www.ibm.com/topics/computer-vision>
- <https://www.coursera.org/learn/introduction-computer-vision-watson-opencv>



What is computer vision ?

The screenshot shows a web browser window with the URL `ibm.com/topics/computer-vision`. The page features the IBM logo and a navigation menu with links: Automation, Strategy and vision, Approach, Technology, and What's next. The main content area has a dark background with a glowing blue circuit pattern. The title 'What is computer vision?' is displayed in large white text. Below it, a paragraph explains: 'Use machine learning and neural networks to teach computers to see defects and issues before they affect operations'. At the bottom left, there is a blue button with the text 'Explore Maximo Visual Inspection' and a right-pointing arrow.

What is computer vision?

Use machine learning and neural networks to teach computers to see defects and issues before they affect operations

[Explore Maximo Visual Inspection](#) →



What is computer vision ?

- Computer vision is a field of artificial intelligence (AI)



What is computer vision ?

- Computer vision is a field of artificial intelligence (AI) that enables computers and systems to **derive meaningful information** from digital images, videos and other visual inputs



What is computer vision ?

- Computer vision is a field of artificial intelligence (AI) that enables computers and systems to **derive meaningful information** from digital images, videos and other visual inputs — and **take actions or make recommendations** based on that information.



What is computer vision ?

- Computer vision is a field of artificial intelligence (AI) that enables computers and systems to **derive meaningful information** from digital images, videos and other visual inputs — and **take actions or make recommendations** based on that information. If **AI** enables **computers to think**, **computer vision** enables them to **see, observe and understand**.



What is computer vision ?

- **Computer vision** works much the **same** as **human vision**, except humans have a head start.



What is computer vision ?

- **Computer vision** works much the **same** as **human vision**, except humans have a head start. **Human** sight has the advantage of **lifetimes** of context to train how to tell objects apart, how far away they are, whether they are moving and whether there is something wrong in an image.



What is computer vision ?

- **Computer vision** **trains** machines to perform these functions, but it has to do it in much **less time** with **cameras, data and algorithms** rather than retinas, optic nerves and a visual cortex. Because a system trained to inspect products or watch a production asset can analyze thousands of products or processes a minute, noticing imperceptible defects or issues, it can quickly surpass human capabilities.



What is computer vision ?

- Computer vision is used in industries ranging from energy and utilities to manufacturing and automotive – and the market is continuing to grow. It is expected to reach USD **48.6** billion by 2022⁽¹⁾

1. <https://www.forbes.com/sites/bernardmarr/2019/04/08/7-amazing-examples-of-computer-and-machine-vision-in-practice/#3dbb3f751018>



Why use computer vision?

Like all technologies,

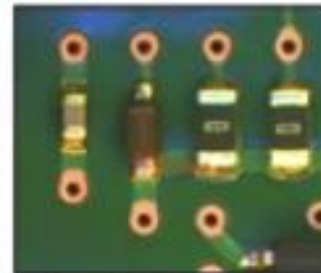
- Slower → Faster
- Expensive → Cheap
- Manual → Automated
- Difficult → Easy
- Inconvenient → Convenient
- Unscalable → Scalable



Computer vision applications

Industries

- Automotive
- Food & Beverage
- Manufacturing
- Pharmaceutical
- Supply chain
- Energy & Utilities
- Hospitality
- Life Sciences
- Human Resources
- Insurance



Computer vision examples

7 Amazing Examples Of Computer And Machine Vision In Practice



Bernard Marr Contributor ©
Enterprise Tech

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<https://www.forbes.com/sites/bernardmarr/2019/04/08/7-amazing-examples-of-computer-and-machine-vision-in-practice/?sh=2f5b2b8f1018>



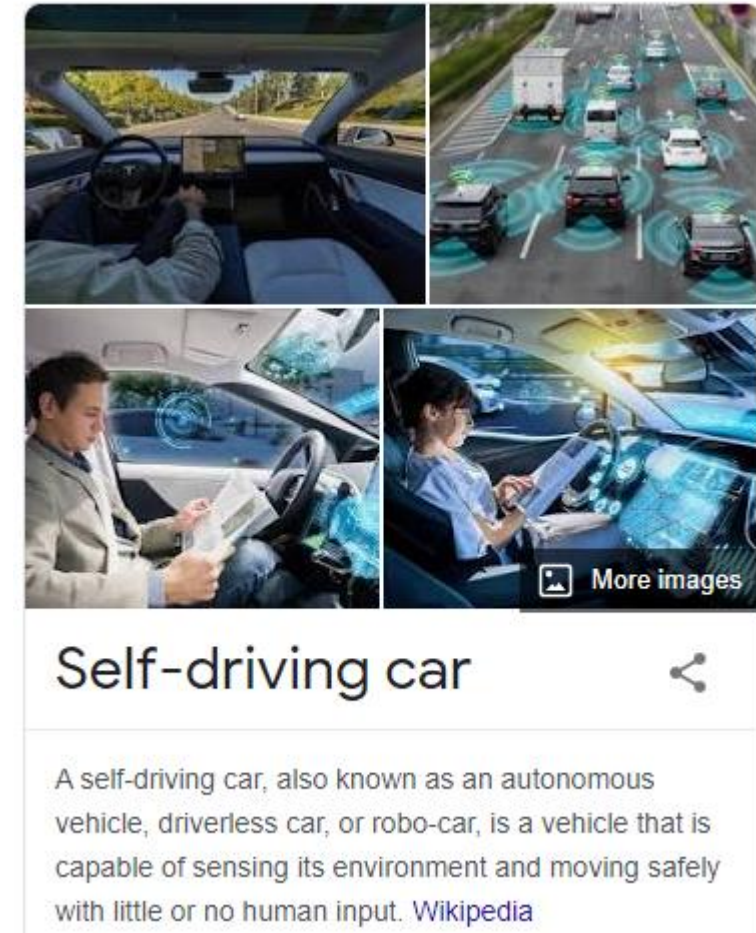
Computer vision examples

1. Autonomous vehicles
2. Google Translate app
3. Facial recognition
4. Healthcare
5. Real-time sports tracking
6. Agriculture
7. Manufacturing

Computer vision examples

1. Autonomous vehicles

Computer vision is necessary to enable self-driving cars. Manufacturers such as Tesla, BMW, Volvo, and Audi use multiple cameras, lidar, radar, and ultrasonic sensors to acquire images from the environment so that their self-driving cars can detect objects, lane markings, signs and traffic signals to safely drive.



Computer vision examples

1. Autonomous vehicles



<https://www.youtube.com/watch?v=tlThdr3O5Qo>

Computer vision examples

2. Google Translate app

All you need to do to read signs in a foreign language is to point your phone's camera at the words and let the Google Translate app tell you what it means in your preferred language almost instantly. By using optical character recognition to see the image and augmented reality to overlay an accurate translation, this is a convenient tool that uses computer vision.



Google Translate 4+
Translate 108 languages
Google LLC
#1 in Reference
★★★★★ 4.5 • 65.8K Ratings
Free



Google Translate

Google LLC Tools

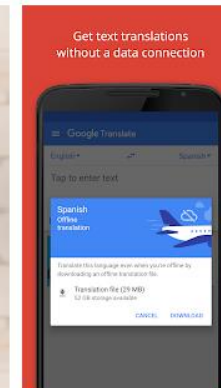
★★★★★ 8,027,274

3+

This app is available for your device

Add to Wishlist

Install



Computer vision examples

2. Google Translate app



<https://mashable.com/article/word-lens-japanese>



<https://www.hellotech.com/blog/the-5-best-free-translation-apps>

Computer vision examples

3. Facial recognition

China is definitely on the cutting edge of using [facial recognition technology](#), and they use it for police work, payment portals, security checkpoints at the airport and even to dispense toilet paper and prevent theft of the paper at Tiantan Park in Beijing, among many other applications.



Computer vision examples

3. Facial recognition

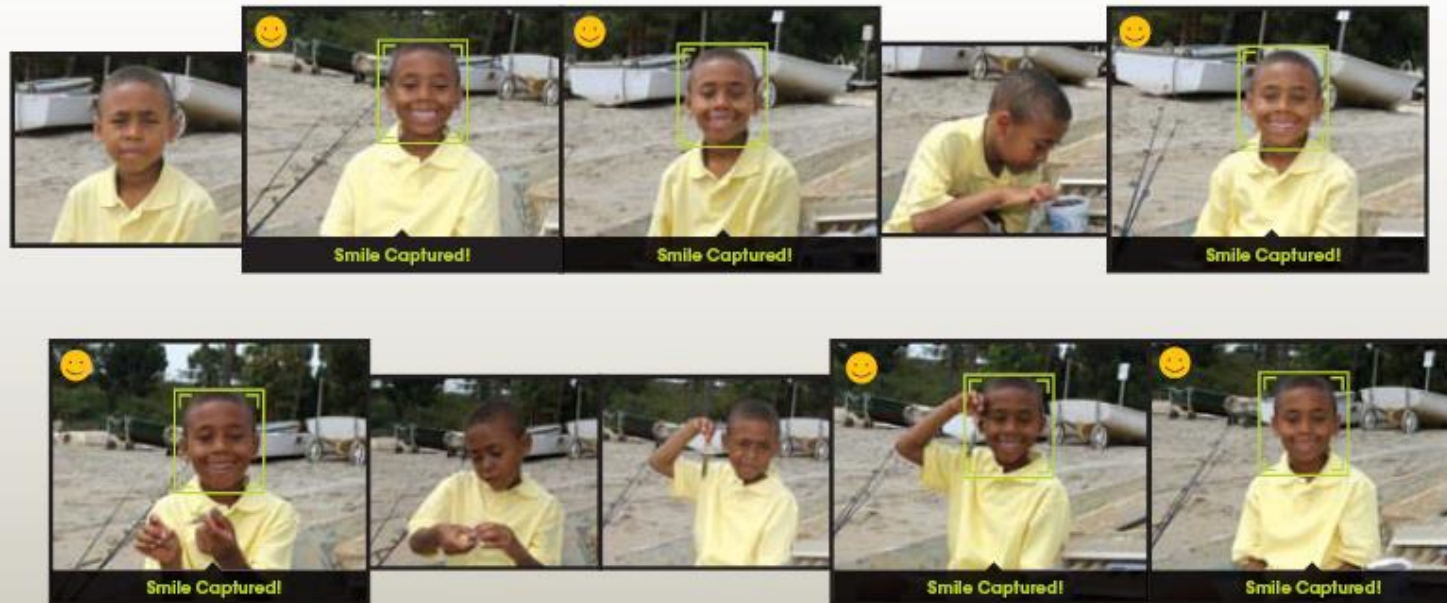


- Many new digital cameras now detect faces
 - Canon, Sony, Fuji, ...

Smile detection

The Smile Shutter flow

Imagine a camera smart enough to catch every smile! In Smile Shutter Mode, your Cyber-shot® camera can automatically trip the shutter at just the right instant to catch the perfect expression.



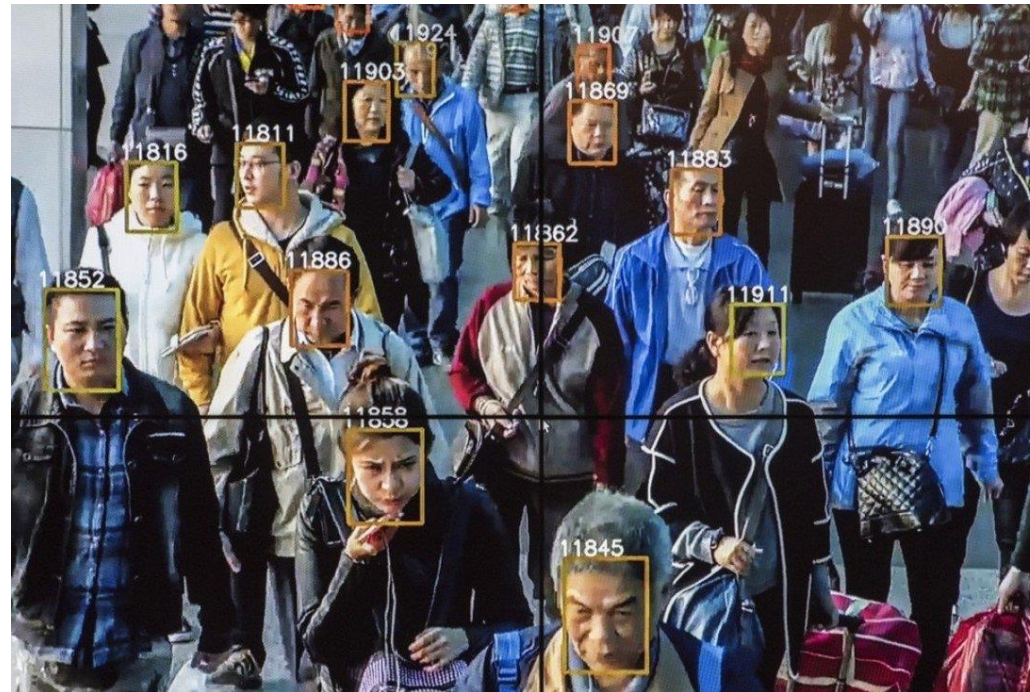
Face recognition: Apple iPhoto, Facebook, Google, etc



Computer vision examples

3. Facial recognition

Shanghai airport first to launch automated clearance system using facial recognition technology



https://www.scmp.com/tech/enterprises/article/2168681/shanghai-airport-first-launch-automated-clearance-system-using?module=perpetual_scroll&pgtype=article&campaign=2168681

Computer vision examples

3. Facial recognition

Alibaba launches 'smile to pay' facial recognition system at KFC in China



An Alibaba employee demonstrates 'Smile to Pay', an automatic payment system that authorize payment via facial recognition

Alex Wong / Staff / Getty Images

Computer vision examples

4. Healthcare

Since [90 percent of all medical data is image based](#) there is a plethora of uses for computer vision in medicine. From enabling new medical diagnostic methods to analyze X-rays, mammography and other scans to monitoring patients to identify problems earlier and assist with [surgery](#), expect that our medical institutions and professionals and patients will benefit from computer vision today and even more in the future as its rolled out in [healthcare](#).



Computer vision examples

4. Healthcare



assisted living, patient monitoring
[Lan et al, PAMI 2012]



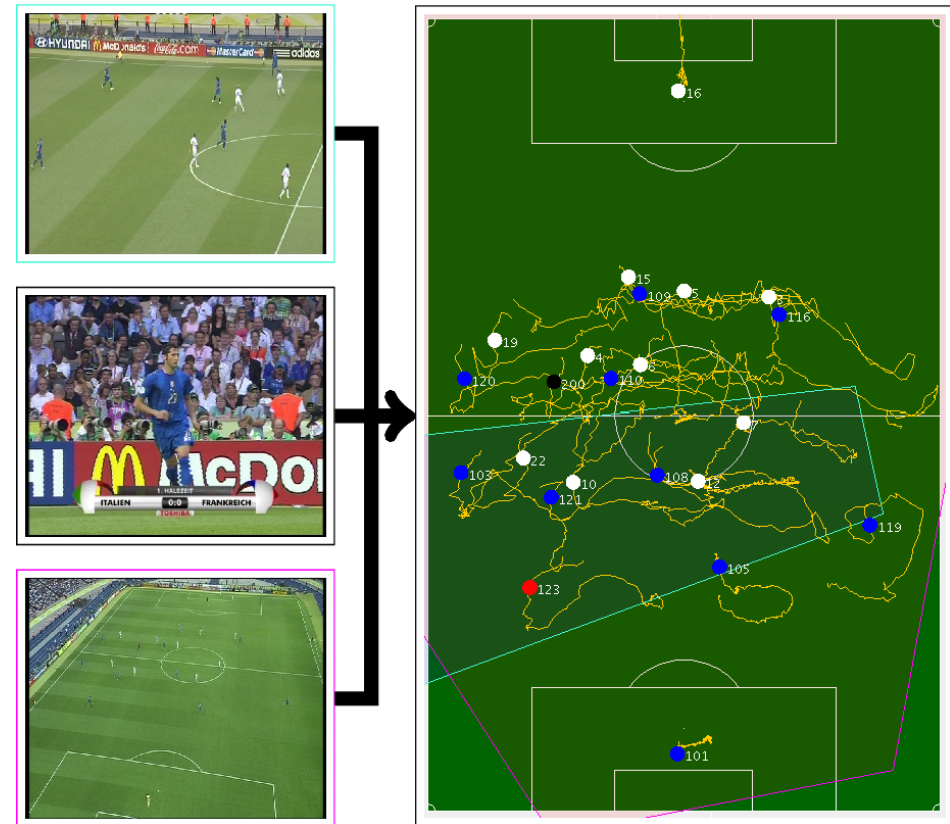
autism screening (sàng lọc tự kỷ)

<http://www.gatech.edu/newsroom/release.html?nid=60509>

Computer vision examples

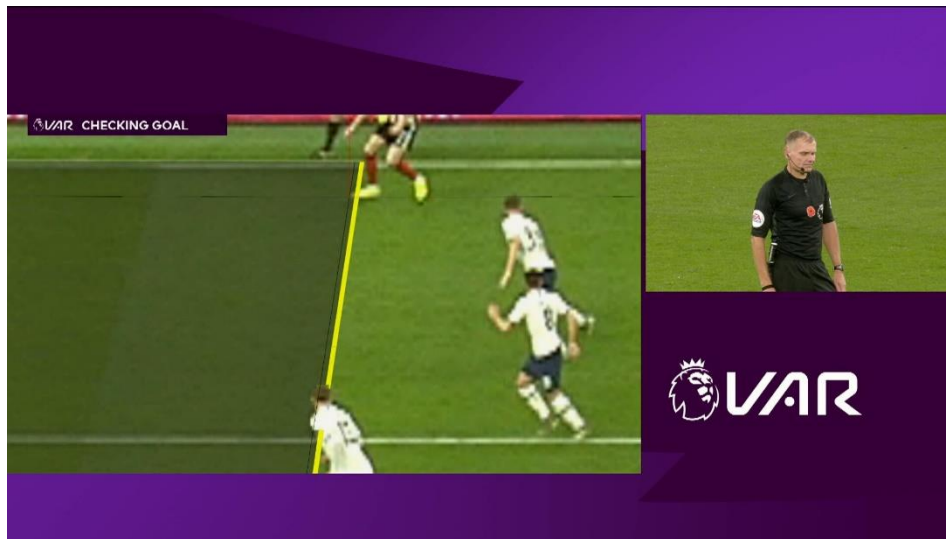
5. Real-time sports tracking

Ball and puck tracking on televised sports has been common for a while now, but computer vision is also helping play and strategy analysis, player performance and ratings, as well as to track the brand sponsorship visibility in [sports broadcasts](#).



Computer vision examples

5. Real-time sports tracking



<https://www.skysports.com/football/news/11095/11928910/var-premier-league-clubs-call-for-change-in-marginal-offside-calls>



Computer vision examples

6. Agriculture

At [CES 2019](#), John Deere featured a semi-autonomous combine harvester that uses artificial intelligence and computer vision to analyze grain quality as it gets harvested and to find the optimal route through the crops. There's also great potential for computer vision to identify weeds so that herbicides can be sprayed directly on them instead of on the crops. This is expected to [reduce the amount of herbicides needed by 90 percent](#).

Computer vision examples

6. Agriculture

×

All

Images

News

Videos

More

Tools

About 139,000,000 results (0.57 seconds)

The agricultural industry has witnessed several contributions of computer vision-artificial intelligence (AI) models in areas such as **planting, harvesting**, advanced analysis of weather conditions, weeding and plant health detection and monitoring. Jan 31, 2021

45

Computer vision examples

computer vision in agriculture - C x +

google.com/search?q=computer+vision+in+agriculture&source=lnms&tbm=isch&sa=X&ved=2ahUKEwjv6-6ewOnyAhXFZSsKHVFWdVvMQ_AUoAXoECAEQAw&bi...

Google computer vision in agriculture

Q All Images News Videos Maps More Tools Collections SafeSearch

robot fruit opencv drone object detection precision agriculture cvpr machine learning crop

Computer Vision In Agriculture. Humans ...
sushantjha8.medium.com

Croptracker - Computer Vision in Agtech
croptracker.com

Machine Vision Saving Agriculture: One ...
automate.org

Application of Computer Vision in ...
medium.com

5 Top Computer Vision Startups in Agriculture
Global Startup Heat Map Inside

Application of Computer Vision in ...

Driving Agriculture into the Future ...

Machine Vision Applications in ...

Computer vision examples

7. Manufacturing

Computer vision is helping manufacturers run more safely, intelligently and effectively in a variety of ways. Predictive maintenance is just one example where equipment is monitored with computer vision to intervene before a breakdown would cause expensive downtime. Packaging and product quality are monitored, and defective products are also reduced with computer vision.

<https://www.devteam.space/blog/10-examples-of-using-machine-vision-in-manufacturing/>

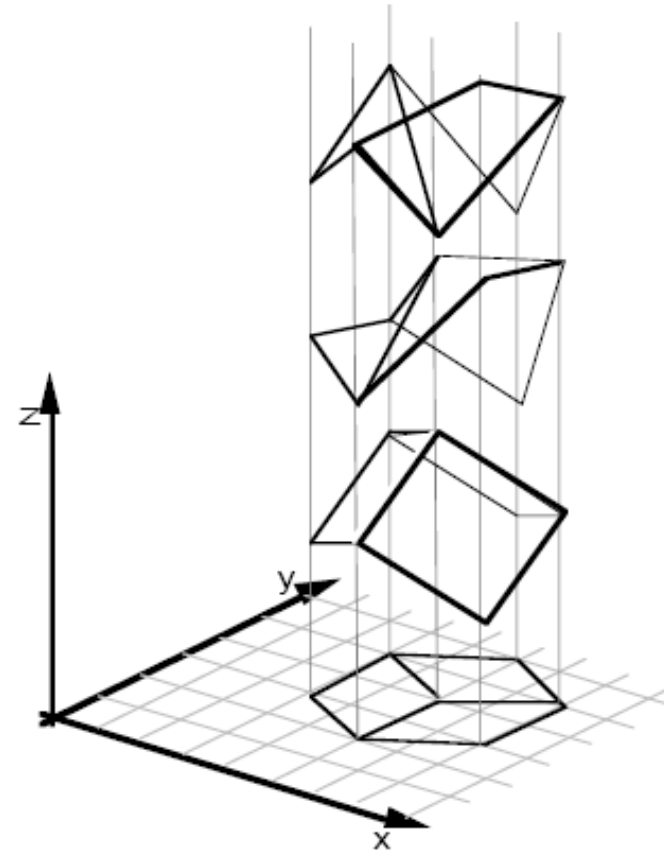




Why vision is so hard?

Why is vision so hard?

- Ill-posed problem



[Sinha and Adelson 1993]

Challenges 1: view point variation



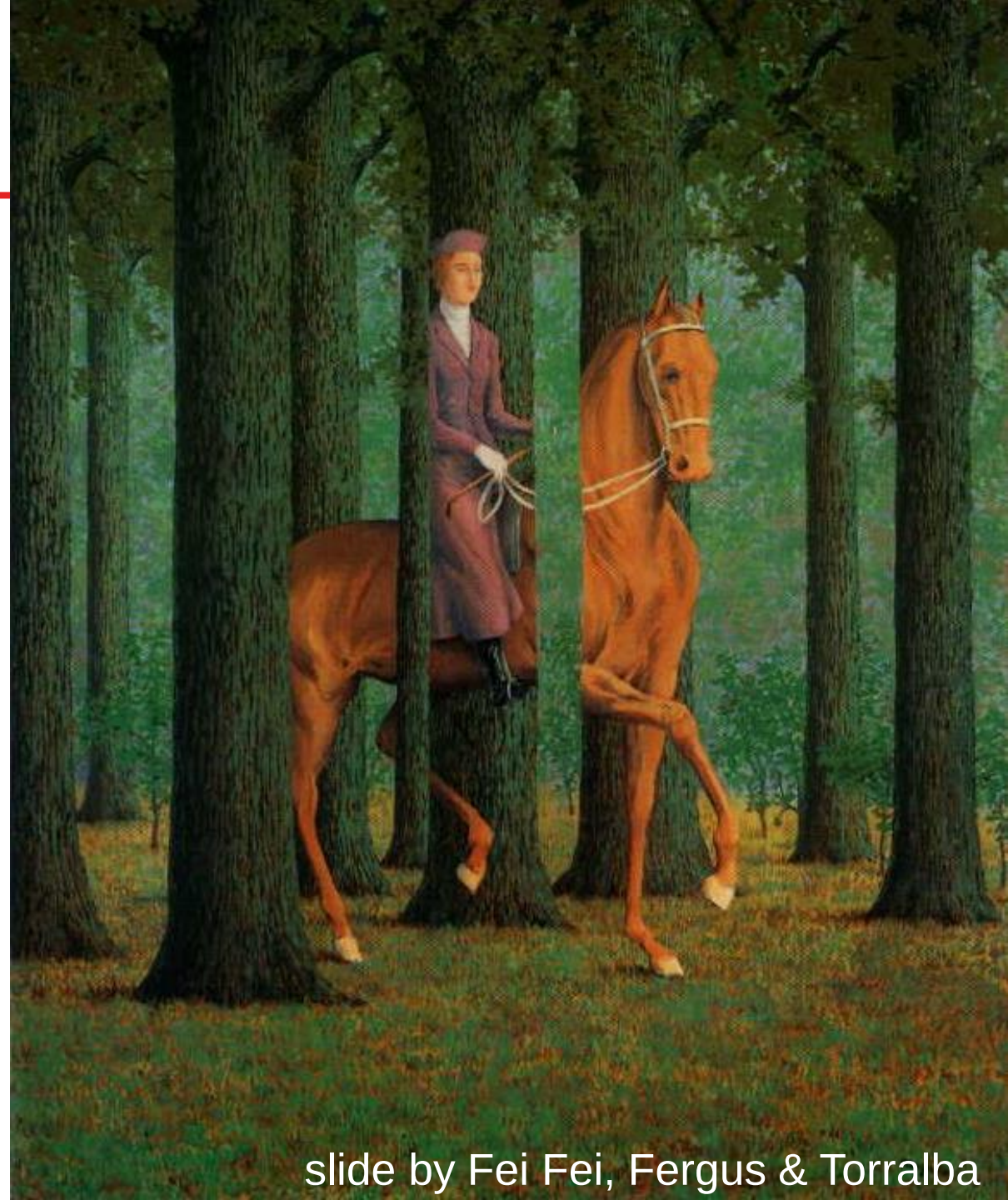
Michelangelo 1475-1564

slide by Fei Fei, Fergus & Torralba

Challenges 2: illumination



Challenges 3: occlusion



Magritte, 1957

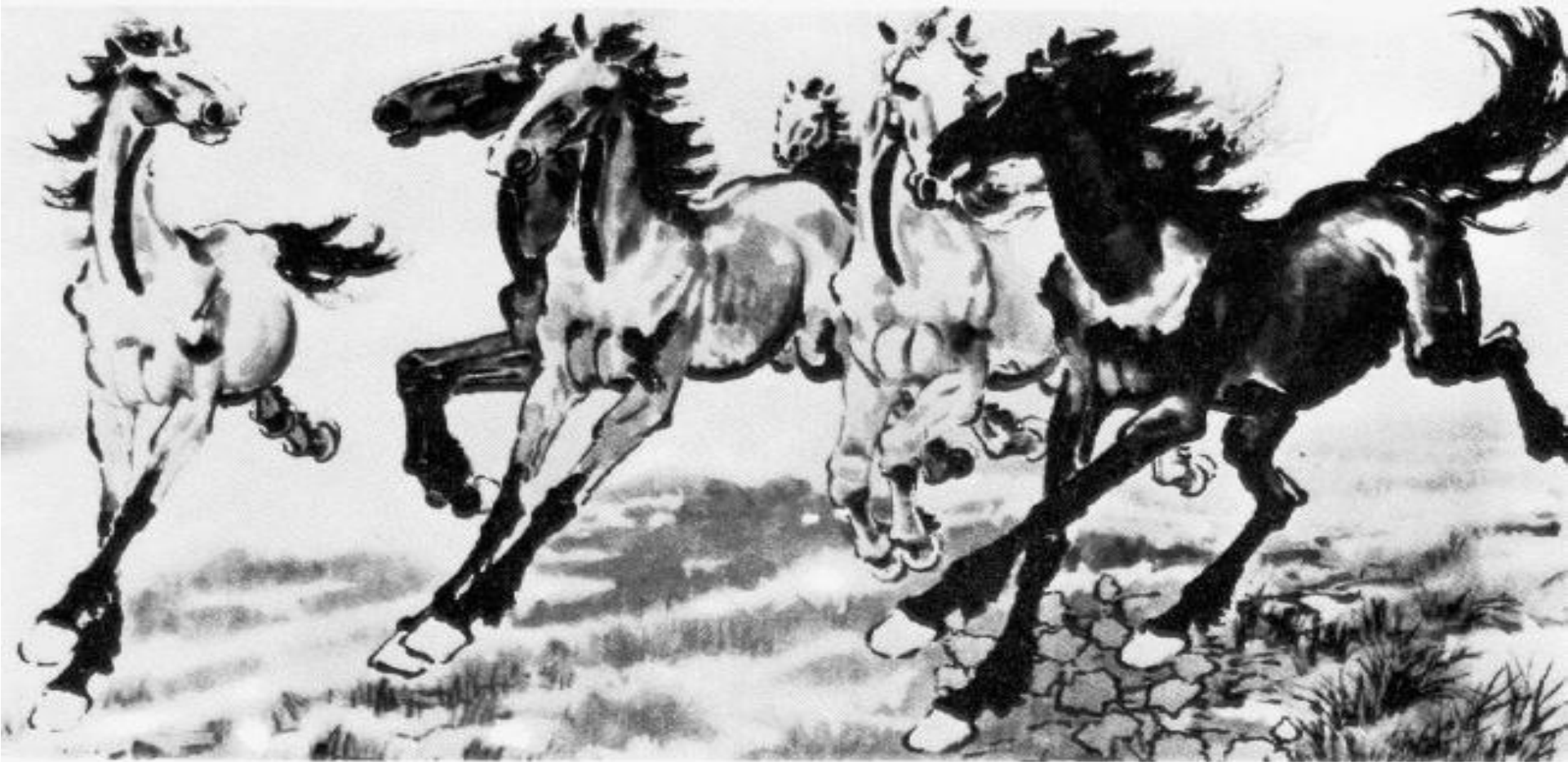
slide by Fei Fei, Fergus & Torralba

Challenges 4: scale



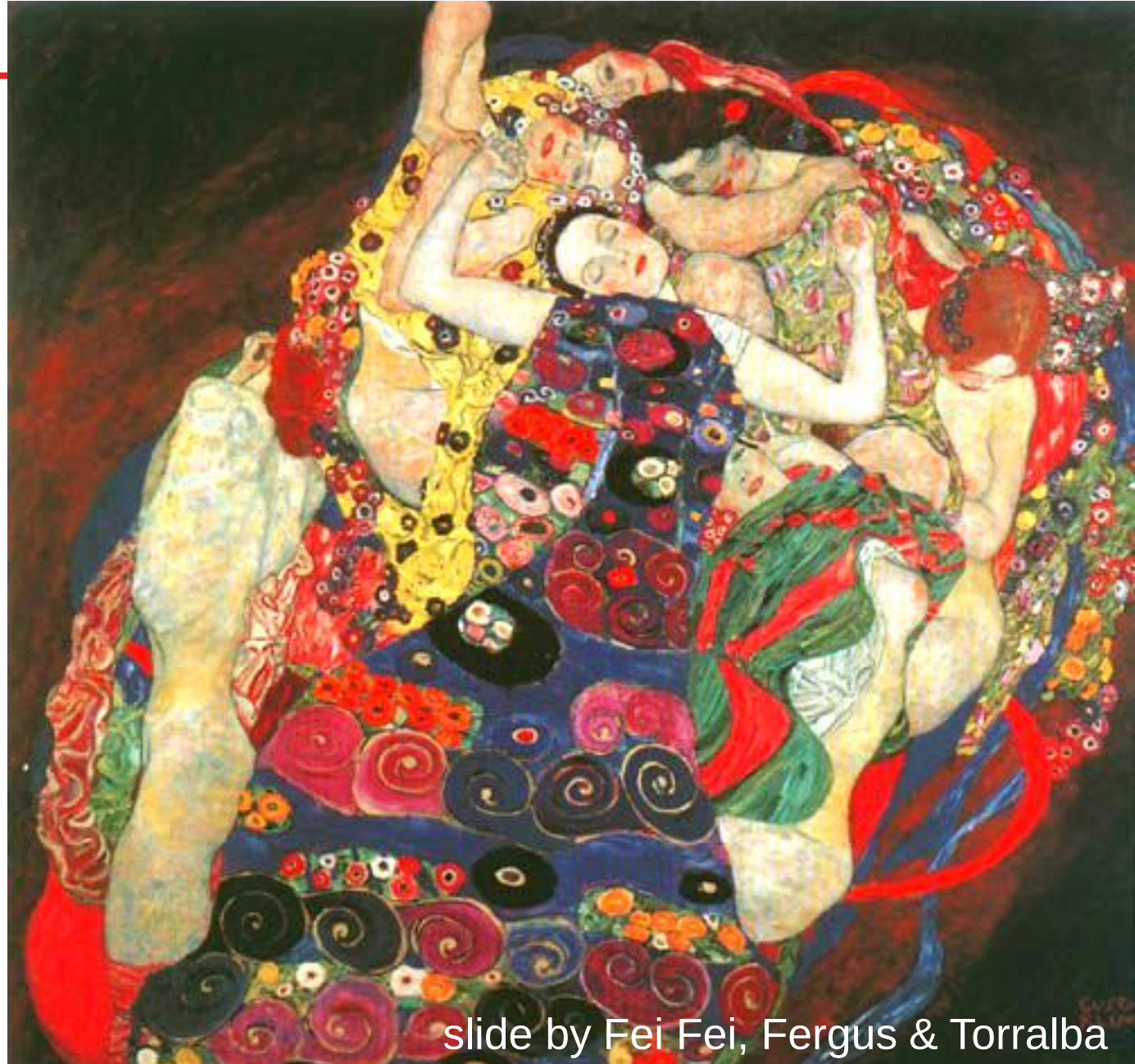
slide by Fei Fei, Fergus & Torralba

Challenges 5: deformation



Xu, Beihong 1943

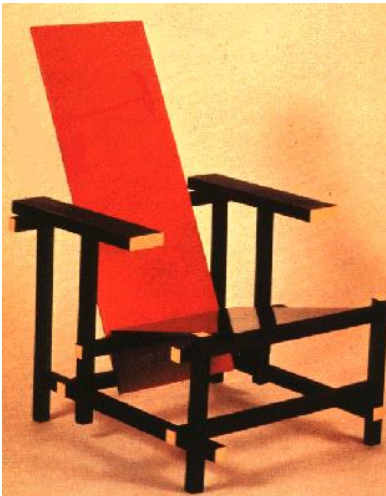
Challenges 6: background clutter



Klimt, 1913

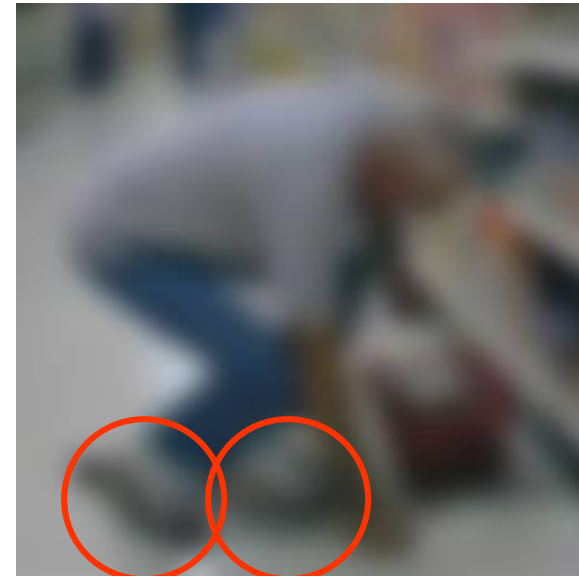
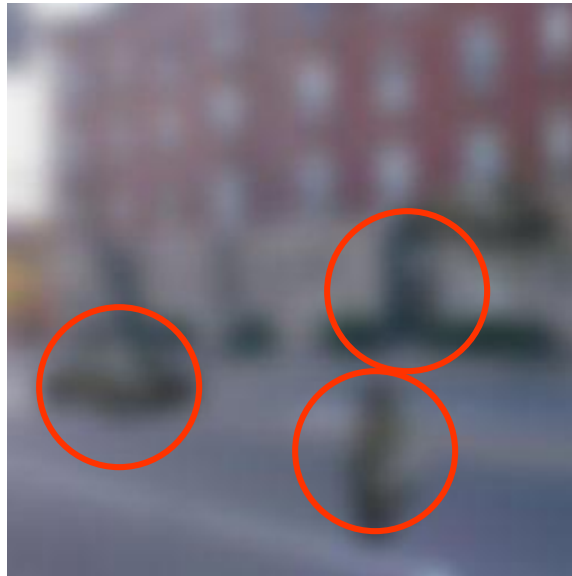
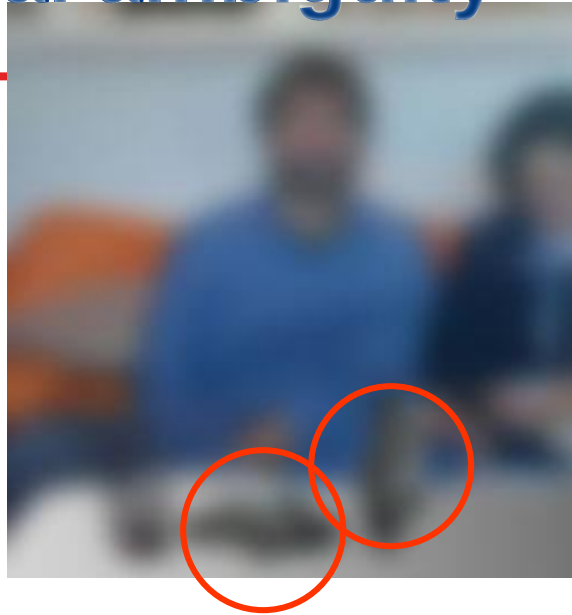
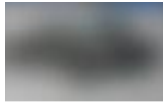
slide by Fei Fei, Fergus & Torralba

Challenges 7: object intra-class variation



slide by Fei-Fei, Fergus & Torralba

Challenges 8: local ambiguity



slide by Fei-Fei, Fergus & Torralba



Challenges 9: the world behind the image





Các bài toán trong thị giác máy tính

- Classification
- Localization
- Object detection
- Segmentation
- ...

Ví dụ

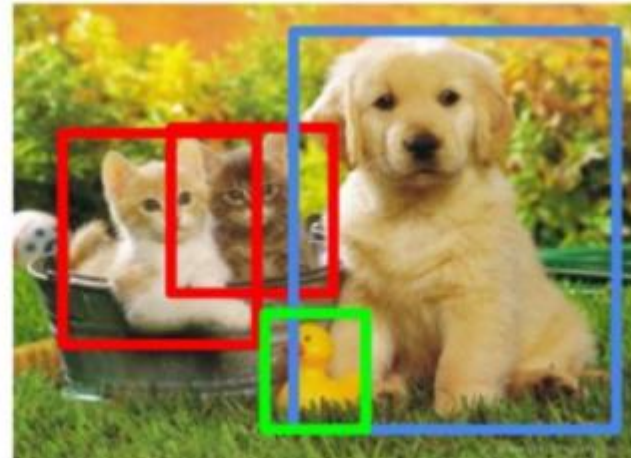
Classification



Classification + Localization



Object Detection



Instance Segmentation

